THE EDITOR OF THE SUITE

bm Mesh: editing meshes

Move the vertices and the faces of a 3D model, on the console, and see it in your game.

Contents

What bm Mesh is 4

What it does 4

How it connects to Studio and Animator 4

Which editor for what 4

The three kinds of meshes 4

Good to know 5

Opening 6

Projects and games 6

Ways to open 6

What you see after opening 6

Caveats 6

The screen and the controls 7

The pages 7

The list page (F1) 7

The edit page (F2) 8

The gamepad 10

Selecting and moving 11

Vertices, edges or faces 11

Move, rotate, scale 11

Other changes 11

Add a part 11

Live mirror 12

Bones 12

Snap 13

Box select, hide and isolate 13

Soft move 13

Type a value 13

Reference pictures 14

Edges and the loop cut 14

Bevel 15

Project onto the sprite sheet . . . 15
Reference model (ghost) . . . 15
The view . . . 16
Caveats . . . 16

The preview and cart_meshes . . . 17
How the game's meshes arrive . . . 17
Good to know . . . 17

Saving . . . 18
Import a .glb . . . 18

Tutorials . . . 19
Move a vertex . . . 19
Fix a face . . . 19
Check the model in a game . . . 20

Tips and limits . . . 21
Tips . . . 21
Limits . . . 21

Open points . . . 22

README

What this is. The manual of **bm Mesh**, the editor of vertices and faces. Every key in it comes from the editor's own code.

How to use it. Read "Opening" and "The screen and the controls" once, then follow the three tutorials with a cartridge of yours.

What is missing. Every gap is marked with a highlighted TO FILL tag; nothing is invented.

What bm Mesh is

bm Mesh is the editor that works on the points (vertices) and on the triangles (faces) of a 3D model. It is built into the console: you find it in the **Dev** tab.

What it does

- It shows the models of a cartridge turning in a preview, with vertices, triangles and skeleton.
- It lets you choose vertices, edges or faces and move, rotate, scale, extrude, mirror, merge, delete and paint them.
- It also reads the meshes that the game's own code builds, so you can start from a ship or a ring made in Lua.

How it connects to Studio and Animator

- **bm Studio** makes the models. bm Mesh edits the same models: they live in the MESH section of the cartridge.
- **bm Animator** adds skeletons and animations (the ANIM section). bm Mesh keeps the skeleton of a model: the bone of each new vertex is the one of the vertex it comes from, and bones and animations stay those of bm Animator. In bm Mesh you can also give vertices to a bone (see Bones).
- In the game you use a model with `model("name")` .

Which editor for what

- **bm Studio**: static models. Objects, props and buildings, made of blocks and tiles. The 3D map of a game is made in the map page of bm SDK.
- **bm Mesh**: organic models. Characters and heroes, and all fine editing of vertices and triangles.
- **bm Animator**: rigs and animations. It puts a skeleton into a model and moves it. It does not build models.
- bm Studio and bm Mesh write the same MESH section, so a model can go from one editor to the other.

The three kinds of meshes

The list marks each mesh with a letter.

Letter	What it is	In the game
M	a model of the MESH section (the ones of bm Studio)	`model("name")`

Letter	What it is	In the game
C	a mesh in the code, written by bm Mesh as `mesh_name()` between the lines `-- [bm Mesh begin]` and `-- [bm Mesh end]` at the end of `main.lua`	`local m = mesh_name()` , then `draw3d(m, ...)`
G	a mesh the game's code builds with `mesh()` , `mesh_sphere()` or `mesh_cube()`	the game's own code

Good to know

- The editor is used with the keyboard, with the mouse or with the gamepad.
- Editing steps can be undone (30 steps): Ctrl+Z undoes, Ctrl+Y redoes (the gamepad: Y + A undoes). Rename and delete clear the undo and redo steps, and so does opening a file; copy as model, copy as code, duplicate and new do not make an undo step.

Opening

Projects and games

All the tools of the suite change only **projects** (`.bme`). A **game** (`.bm` or `.b16`) they only read.

- If you open a game, bm Mesh reads it and you can edit. The first save makes an editable copy, named after the game with the extension `.BME` , and from then on you work on the copy.
- The status bar tells you when you opened a game: `a game: saving makes its copy` .

Ways to open

- From the **Dev** tab of the console menu.
- From a game: press **X** on its cover, then **Open in bm Mesh**.
- From the monitor: the key `4` .
- From inside the editor: Esc, then **Open...** (Ctrl+O). The list shows the `.bme` projects of `/bme` and the `.bm` and `.bme` files of `/carts` and of the root of the SD card.
- If you started bm Mesh with no file, the file chooser opens by itself.

What you see after opening

The status bar says how many models, code meshes and meshes from the game's code were found, and the editor shows the **list** page.

Caveats

- If the game's code stops while it is read, you get the message `the game's code stopped while read` : the meshes found before the error are still there.
- A file that cannot be opened gives `cannot open ...` with the reason.

The screen and the controls

The pages

The tab bar at the top starts with the name `bm Mesh`. Then it has three pages. A click on a tab, or its key, changes page.

Page	Key	What it is for
list	F1	choose a mesh; copy, rename, duplicate, delete, make a new one
edit	F2	choose and change the vertices and the faces
menu	Esc	open, save, try the game, exit

[TO FILL: shots/editor-mesh-list.png]

Other keys. F3 works only on the list and edit pages, and `[` and `]` are ignored on the menu page; the others work on every page:

Key	What it does
F3	the commands of the page, as a menu
`[` and `]`	the previous and the next mesh
F12 held, or `?`	shows the keys of the page
Ctrl+S	save
Ctrl+Shift+S	save as (a name of 8 letters, in `/carts`)
Ctrl+O	open another file
Ctrl+N	a new model, a cube
Ctrl+I	import a model from `/bmm` or `/glb`
Ctrl+E	export the mesh to `/bmm`
F5 or Ctrl+R	try the game
F6	the assistant
Ctrl+Z, Ctrl+Y	undo, redo
Ctrl+Esc	back to the bm menu (it asks first if there are unsaved changes)

The list page (F1)

Key	What it does
up, down, home, end	choose the mesh
Enter	edit it

Key	What it does
m	copy as a model (mesh to model)
c	copy as code (model to `mesh_name()`)
r	rename
d	duplicate (a model keeps its skeleton)
Del	delete: press it twice
n	a new model, a cube; with F3 also a plane, a sphere, a cylinder or the humanoid base
i	import a model from `/bmm` or `/glb`
x	export the mesh to `/bmm`
q, e	turn the view
w, s	tilt the view
+ and -	zoom
space or o	start or stop the spin
f	frame the mesh again

A G mesh cannot be renamed or deleted: the game's code makes it. Copy it with **m** or **c** to get one you can name.

The edit page (F2)

The edit page draws the model with its edges, a pointer, and a line at the top with the name, the number of vertices and triangles, the mode (`VERTICES` or `FACES`) and how many are chosen. A small colour swatch and the current step are shown under it.

Key	What it does
arrows	move the pointer (faster when held)
home	the pointer to the middle
n, b	the pointer on the next or the previous vertex or face, also one behind
space	add to the choice, or take away
Enter	choose only that one
a	choose all; if something is chosen, choose none
l	choose everything connected to it
Tab	switch between vertices, edges and faces
g, r, t	move, rotate, scale
x	extrude the chosen faces

Key	What it does
d	duplicate the chosen faces
A (Shift+A)	add a part: a cube, a cylinder, a sphere or a plane
m	mirror the choice, left to right on the screen
M	copy the choice to the other side of 0
X (Shift+X)	live mirror on or off
B (Shift+B)	give the chosen vertices to a bone of the skeleton
v	bone view: the vertex dots take the colour of their bone
h	show or hide the ghost (the reference model)
H (Shift+H)	load a ghost from `/bmm` or `/glb`, or clear it
[and]	the ghost smaller or larger (5 percent each press)
{ and } (Shift+[and Shift+])	the ghost down or up (0.05 each press)
S (Shift+S)	snap: off, grid, vertex, grid and vertex
G (Shift+G)	grid step: 0.05, 0.1, 0.25, 0.5
z	box select: the pointer is one corner; move it to the other corner, Enter chooses only what is inside, Space adds it, Esc cancels
y	hide the chosen vertices; with nothing chosen it shows everything again
Y (Shift+Y)	isolate: hide everything but the choice; with nothing chosen it shows everything again
P (Shift+P)	soft move: radius off, 0.25, 0.5, 1, 2 (also inside a move)
R (Shift+R)	reference pictures: a front and a side picture from `/pics`
Ctrl+R	loop cut across the chosen edge
Shift+J	knife: cut a triangle between the middles of two chosen edges
Shift+E	bevel: each chosen edge becomes a narrow strip (you type the width)
Shift+U	project the faces onto the sprite sheet (you type cell size, x and y)
j	make a face on 3 or 4 chosen vertices
k	merge the chosen vertices into one
K	weld the vertices at the same point
u	split each chosen face into 4
i	turn the chosen faces over

Key	What it does
Del	delete the choice
p	paint the chosen faces with the current colour
o	take the colour of the face under the pointer
c	open the palette
q, e	turn the view
w, s	tilt the view
+ and -	zoom
f	frame the chosen vertices
1, 3, 7, 0	front view, side view, top view, the free view
`,` and `.`	a finer or a coarser step

[TO FILL: shots/editor-mesh-edit.png]

The gamepad

Control	What it does
Y + left or right	change page
Y + up or down	the previous or the next mesh
Y + B	the menu
Y + A	undo
X, tapped	the commands of the page
X + d-pad	turn and tilt the view
X + A, X + B	zoom
A	in the list: edit the mesh; in the edit page: choose the vertex or face under the pointer
B	in the edit page: choose none
d-pad	the pointer; inside a tool, change the value

[TO FILL: the pad controls are written from the code; not tried on the PowKiddy RGB30]

Selecting and moving

Vertices, edges or faces

The edit page has three modes. **Tab** goes from one to the next: vertices, edges, faces, and again vertices. The line at the top says `VERTICES`, `EDGES` or `FACES`. In each mode a choice is a set of vertices, a set of edges or a set of faces.

- Put the pointer on a vertex, on an edge or on a face: it lights up. An edge is the line between two vertices of a triangle; the pointer must be within 6 pixels of it.
- **space** adds it to the choice, or takes it away if it was in. **Enter** makes it the only one.
- With the mouse, a click chooses only what is under the pointer; Shift + click adds or takes away.
- In a dialog line (for example the file name), a click puts the caret between the letters.
- **a** chooses all; press it again to choose none. **l** adds everything joined to the choice by faces.
- **n** and **b** walk through all the vertices, edges or faces in order: this reaches also the ones hidden behind the model.

When you switch mode, the choice follows: from vertices to edges, the edges whose two ends are chosen; from edges to faces, the faces whose three corners are chosen; from faces to vertices, the corners of the chosen faces.

Move, rotate, scale

Press **g** to move, **r** to rotate, **t** to scale the choice (the corners of chosen faces move too). The tool opens and the top line shows the value, for example `MOVE x 0 y 0.25 z 0`.

- The arrows and PgUp/PgDn change the value. For a move: left and right go along the view's horizontal axis, up and down change the height, PgUp and PgDn go nearer or farther.
- **x**, **y**, **z** limit the change to that axis. **n** works along the normal. Press the same key again to free it.
- `,` and `.` change the step. The move steps are 0.01, 0.05, 0.1, 0.25, 0.5 and 1; the rotation steps are 1, 5, 15, 30, 45 and 90 degrees; the scale steps are 1.01, 1.05, 1.1, 1.25, 1.5 and 2.
- Type **0** to **9** to enter the exact value (see *Type a value*). **Shift+P** turns the soft move on (see *Soft move*).
- **Enter** confirms. **Esc** cancels and puts everything back.
- With the mouse, drag the left button to change the value and click to put it. With the gamepad, A confirms, B cancels and a tap of X switches the axis.

Other changes

- **x** extrudes the chosen faces; they then move along the normal.
- **j** makes a face on 3 or 4 chosen vertices, in the order you chose them, turned towards the camera. It works in vertex mode only.
- **k** merges the chosen vertices in their middle; **K** welds the vertices that are at the same point (with flag 4 of `draw3d` the faces around look smooth).
- **Del** deletes: in vertex mode the vertices and the faces that use them, in face mode the faces.
- **p** paints, **o** takes a colour, **c** opens the palette (arrows to choose, Enter to take it, Esc to close). A face with a texture of the sheet has no colour.

Add a part

Shift+A opens a list of four shapes: cube, cylinder, sphere and plane. The one you pick is added to the model as a new part, with the width 1 and the colour of the page. It goes to the middle of the chosen vertices, or to 0 if nothing is chosen. The vertices of the new part are then the choice, so you can move, rotate or scale it at once. With the live mirror on (Shift+X), the mirror image across X = 0 is added too, and all the new vertices are the choice. One undo takes away both parts. If the model has a skeleton, the new vertices follow the bone of the first chosen vertex (or the first bone). With the gamepad, tap X and pick **Add a part**.

Live mirror

Shift+X turns the live mirror on and off. While it is on, the page shows **MIRROR X**. Move, rotate, scale, paint, delete, add a part and the loop cut then act on the chosen vertices or faces and on their counterparts across the plane X = 0. The editor finds the counterpart by position: the vertex at (-x, y, z), within 0.001. A vertex on the plane is its own counterpart. A vertex with no counterpart changes alone. If you choose both sides, the vertex with the lower number leads and the other follows. Two tools that change the topology work on both sides. Shift+A adds the part and its mirror image: the second part is the exact mirror copy, with the faces still looking out, and it takes the mirror bone. If the middle of the part is on the plane (x within 0.001 of 0), the part is added once. Ctrl+R cuts the ring of the chosen edge and the ring of its counterpart edge. If the counterpart edge does not exist, one ring is cut and a message tells it. If the ring crosses the plane and holds the counterpart edge, it is cut once. In edge mode, choosing an edge also chooses its counterpart, and Ctrl+R accepts that pair. The other topology tools (extrude, duplicate, subdivide and the knife) do not act on the other side. The sphere and the cylinder have triangles that are mirror images across X = 0, so every vertex that Subdivide adds has a counterpart. One undo takes back both sides. With the gamepad, tap X and pick **Live mirror X on / off**.

Bones

A model with a skeleton gives each vertex to one bone. bm Animator makes the bones. In bm Mesh you choose which vertices follow which bone.

1. Choose vertices or faces. To choose a whole part, choose one vertex of it and press **I**.
2. Press **Shift+B**. A list shows the bones with their numbers and names.
3. Pick a bone and press Enter. The chosen vertices (the corners of the chosen faces) now follow that bone. A message tells how many.

The header shows the bone of the choice on the right of the second row, for example **BONE 3 arm.L**. If the choice has vertices of several bones, it shows **BONES 3: arm.L+** (the number of bones and the first one). With nothing chosen, it shows the bone of the vertex under the pointer. **v** turns the bone view on and off: the vertex dots take the colour of their bone (12 colours, then they repeat). One undo takes back the whole change.

With the live mirror on, the counterparts across X = 0 follow too. If the skeleton has a mirrored pair by name, the counterparts take the other bone of the pair: `arm_l` and `arm_r`, `hand.L` and `hand.R`, `l_foot` and `r_foot`, `armLeft` and `armRight`. A bone with no pair in the skeleton gives the same bone to the counterparts. If you choose both sides, the vertices on the side of the head of the bone keep that bone and the other side takes the pair. A vertex on X = 0 keeps the bone you picked.

If a skeleton has no vertices yet, all vertices are on bone 0 until you give them another bone. If the skeleton was made for a different number of vertices, bm Mesh does not change the bones: skin the model again in bm Animator. A model with no skeleton answers "no skeleton". With the gamepad, tap X and pick **Give the choice to a bone** or **Bone view**.

Snap

Shift+S changes the snap. The order is: off, grid, vertex, grid and vertex, then off again. The page shows the mode at the top right, for example **SNAP grid 0.1**, and a message tells the new mode. The snap works while you move vertices (**g**, and the move after **d**). **Shift+G** changes the grid step: 0.05, 0.1, 0.25 or 0.5 units. You can press both keys while a move is open.

The snap works on the leading vertex. This is the chosen vertex with the lowest number. The editor snaps the position of this vertex, and every other chosen vertex moves by the same distance. The grid mode rounds each axis of the leading vertex to the step. If you lock an axis (x, y or z), only that axis is rounded. The vertex mode moves the leading vertex to the nearest other vertex of the model when that vertex is within 6 pixels on the screen. A small green ring shows the vertex that the snap took. The chosen vertices are never a target. In the mode grid and vertex, a vertex in range wins, and the grid works when no vertex is in range. With an axis locked, the vertex snap changes only that axis.

With the live mirror on, the counterparts follow the same move, mirrored. One undo takes back the whole move. The tool line shows **SNAP** and the move after the snap. The snap does not change a rotate or a scale, and it does not place a new part. With the gamepad, tap X and pick **Snap: off, grid, vertex, both** or **Snap grid step**.

Box select, hide and isolate

Box select. Press **z**. The pointer is now one corner of a box. Move the pointer with the arrows to the other corner: the box is drawn. **Enter** chooses only what is inside the box, **Space** adds it to the choice, **Esc** cancels. With the mouse, drag the left button over the model: when you let go, what is inside is chosen; hold Shift to add. The box takes everything inside it, also the vertices behind the model. In vertex mode it chooses the vertices inside. In edge mode it chooses the edges whose two ends are inside. In face mode it chooses the faces whose three corners are inside. With the gamepad, tap X and pick **Box select**: the A button chooses, the B button cancels.

Hide and isolate. Press **y** to hide the chosen vertices, and the faces that use them. Press **Shift+Y** to isolate: everything but the choice is hidden. The top right shows how many vertices are hidden, for example `12 HIDDEN`. A hidden vertex is not drawn, not under the pointer, not chosen by **a**, **l** or a box, and no tool moves it. With the live mirror on, the counterparts are hidden too. Press **y** (or **Shift+Y**) with nothing chosen to show everything again. The hidden vertices are still in the model: the file keeps all of them, and the counters show all of them. Hiding is part of the undo steps, so **Ctrl+Z** shows them again. The hidden set is forgotten when you change to another mesh.

Soft move

Shift+P turns the soft move on. It cycles the radius: off, 0.25, 0.5, 1 and 2 units. The top right shows it, for example `SOFT 0.5`. With the soft move on, a move (**g**) also pulls the vertices near the chosen ones: a vertex at the radius does not move, a vertex next to the choice follows almost fully, and the fall in between is smooth. The distance is the distance in space to the nearest chosen vertex. The tool line shows the radius and how many vertices follow, for example `soft 0.5 (34 near)`. You can press **Shift+P** inside the move to change the radius. The soft move acts on a move only, not on a rotate or a scale. Hidden vertices do not follow. With the live mirror on, each counterpart pulls its own neighbours, mirrored. The snap and the typed value work with it. One undo takes back the move and all the vertices it pulled.

Type a value

In a move, a rotate or a scale you can type the exact value. Press **x**, **y**, **z** or **n** to choose the axis, then type the number with the keys **0** to **9**, **.** and **-** (the minus first). The top line shows `typed 1.25_` and the model follows at once. **Backspace** takes the last character off. For a move the number is the distance in units on that axis. For a rotate it is the angle in degrees. For a scale it is the factor. If you chose no axis, a move goes along x, a rotate turns around y, and a scale changes all three axes. Press another axis key to type a value for that axis too: `x 3` , then `y 4` gives a move of 3 along x and 4 along y. The typed value replaces the arrow steps of that axis. **Enter** confirms and **Esc** cancels, as always. Start a number below 1 with the zero (`0.5`): the point alone is the step key. Typing needs a keyboard.

Reference pictures

A reference picture is a front or a side picture that stands behind the model while you edit. Put the pictures in the `/pics` folder of the SD card (PNG, baseline JPEG or WebP).

1. Press **Shift+R**. A list shows **Front picture** and **Side picture**. On the gamepad, tap X and pick **Reference pictures**.
2. Pick one. A second list shows the pictures of `/pics` . Pick one.
3. Press **1** for the front view, or **3** for the side view. The picture is drawn behind the model, flat on the plane through the middle of the model, with the model's feet at its lower edge.

The picture shows when the view looks straight at it, within about 17 degrees. The front picture shows from the front and from the back (mirrored), the side picture from both sides. In other views nothing is drawn. The picture is made to fit 256 by 256 pixels and is drawn pale, so the model stays easy to see. The list of **Shift+R** also has **Hide the pictures**, **Bigger**, **Smaller**, **Up**, **Down** and **Clear the pictures**. At first the picture is as high as the model. Use Bigger and Smaller until the picture fits, and Up and Down to put the feet on the floor.

The pictures are never saved in the model or in the project. They live in the sprite sheet, below the rows of the project: two places of 256 rows each (512 rows). Clear puts the sheet back to its own size. The export to `/bmm` writes only the rows of the project. Opening another project forgets the pictures. The Raspberry Pi Zero W has little memory: two small pictures cost about 260 KB in the sheet. Use pictures not larger than 1000 pixels: the console reads the whole file first, then makes it smaller.

Edges and the loop cut

In edge mode a choice is a set of edges. The chosen edges show as thick lines. Move, rotate and scale act on the two ends of each chosen edge. **x** and **d** act on the faces whose corners are all chosen. **Del** removes the triangles next to the chosen edges, and the vertices that no triangle uses any more. With the live mirror on, choosing an edge also chooses its counterpart across X = 0, and move, rotate, scale and delete act on both. The loop cut and the knife do not act on the other side.

Ctrl+R makes a loop cut. Choose one edge first (in vertex mode, two vertices that make an edge are enough). The editor goes from the edge across the quads around the model. A quad is two triangles that look the same way (the angle between their normals is less than 60 degrees) and share a diagonal. In each quad it crosses the edge and the opposite edge, and goes on from that edge. It puts one new vertex in the middle of each crossed edge and cuts each quad in four triangles. A new vertex takes the bone of the lower end of its edge and the mean of the texture points (UV). A new triangle takes the colour of the old face that holds most of its corners.

The ring stops in two cases. At an open border (an edge with one triangle only) there is nothing to cross. At a triangle that is not part of a quad, the triangle is cut in two, from the new vertex to its far corner, so no gap stays. The ring is closed if it comes back to the first edge. If the ring would cross itself, or an edge has more than two triangles, nothing changes and a message tells why.

When it is done, the new vertices are the choice (in edge mode, the new edges across the quads), so you can move or scale the loop at once. With the live mirror on, the ring of the counterpart edge is cut too, and the new vertices of both rings are the choice. One undo takes back the whole cut, both rings. With the gamepad, tap X and pick **Loop cut on the chosen edge**.

Shift+J is the knife. Use it in edge mode. Choose two edges of one triangle (two edges that meet at a corner of the triangle). The editor puts a new vertex in the middle of each edge and cuts the triangle so that a new edge joins the two new vertices. The triangle across each of the two edges is cut in two, from the new vertex to its far corner, so no crack appears. A new vertex takes the bone of the lower end of its edge and the mean of the texture points (UV). A new triangle keeps the colour of its old face. The two new vertices are the choice (the new edge, in edge mode), and one undo takes back the whole cut.

The knife does nothing, and a message tells why, in these cases: you did not choose exactly two edges, the two edges are not edges of one triangle, or an edge has more than two triangles. The knife works on one triangle, so the live mirror does not copy it to the other side. With the live mirror on, choosing an edge also chooses its counterpart, so turn the mirror off (Shift+X) before you choose the two edges. Cut the other side by hand. With the gamepad, tap X and pick **Knife on two edges of a triangle**.

Bevel

Shift+E bevels edges. Use it in edge mode. Choose the edges, press **Shift+E** and type the width (the first value is 0.1). In vertex or face mode the edges with both ends chosen are used. Each chosen edge becomes a narrow strip of two triangles, and the hard corner gets a soft highlight.

The two faces of an edge move away from it by the width, and the strip joins them. At the end of an edge the corner vertex is split in two (or in more, where several chosen edges meet; three edges or more get a cap face), and one face is cut in two, so no gap stays. The new faces keep the colour, the sheet uv and the bone of the faces and vertices they come from.

- A flat edge (two faces that look the same way) is left out, and the message counts it.
- The width is cut to 0.45 of the smallest face height over a chosen edge, so the strip cannot fold over. The message says when it was cut.
- The editor refuses an edge on an open border or with more than two faces, and a vertex whose faces do not close round it. A message tells which, and nothing changes.
- When the strips are done, they are the choice (face mode). One undo takes back the whole bevel.
- With the live mirror on, the counterparts across X = 0 are bevelled too. Both sides then get the same width.

Project onto the sprite sheet

Shift+U gives a place on the sprite sheet to each chosen face (or to every shown face when none is chosen). This is box projection. Type the cell size, the x and the y of the top left corner, for example `32 0 0`. The editor uses 6 cells of that size, 3 across and 2 down: +x, -x, +y on the first row, and -y, +z, -z on the second. Each face goes to the cell of the side it looks to, drawn as seen from that side. All the cells have the same scale, so the texel size is the same on every side. One pixel at the edge of each cell stays free.

The faces become faces with a texture of the sheet. The editor does not paint the sheet: paint the cells in bm Pixel, then look at the model. If the sheet is too small for the 6 cells, a message says so and nothing changes. The faces that were flat colours lose their colour. One undo takes the projection back.

Reference model (ghost)

A ghost is a model that you show as a faint reference while you edit another model. You trace over it. It is drawn as a thin green wire behind your work. It is not part of your model: you cannot choose it, it is not in the vertex and triangle counters, and it is never saved in the project or in a model. The live mirror and the snap do not act on it.

1. Open a model on the edit page. Press **Shift+H**. On the gamepad, tap X and pick **Ghost: load a model, clear it**.
2. Choose a model from `/bmm` or `/glb`. A `.glb` is only read: bm Mesh does not convert it and does not write anything to `/bmm`. The status line tells the name and the counts, for example `ghost robot.glb: 1200 vertices, 2000 triangles`. The page shows **GHOST** and the name at the top right. The last row of the chooser is **Ghost detail**. Press Enter on it to change the detail: 2000, 5000, 20000 or all faces. The detail applies to the next load. If a ghost is loaded, bm Mesh reads it again with the new detail and keeps its scale and height. The detail is a setting of the editor: it is not saved in the model.
3. Press **h** to hide the ghost and **h** again to show it. The ghost stays when you change page or model.
4. Make the ghost fit your work. **[** and **]** make it smaller and larger. **{** and **}** (**Shift+[** and **Shift+]**) move it down and up. The scale goes from 0.1 to 10. The ghost uses the same camera as your model, so it stays aligned when you turn the view.
5. To remove the ghost, press **Shift+H** and choose **Clear the ghost**. The memory of the ghost is freed.

With a ghost loaded on the edit page, **[** and **]** change the ghost scale and no longer change to the previous or next mesh. Use the list page (F1) to change the mesh.

The ghost has a budget of faces. The default is 20000. A model with more faces shows only part of them: a `.glb` is reduced by the console, and a `.bmm` shows 1 triangle in 2, 3 or more. The status line says it when this happens. The ghost has a second limit: one frame draws at most 4000 edges of the ghost. When the wire has more edges, bm Mesh draws 1 edge in 2, 3 or more, and the set of edges that is drawn moves on each frame. The wire then looks thinner and can flicker a little. The status line says it, for example `each frame draws 1 edge in 8 (the line budget)`. Choose a lower detail to get a steadier wire. These limits are for the picture only: the file is not changed. The ghost has no texture.

The view

q and **e** turn, **w** and **s** tilt, **+** and **-** zoom, **f** frames the choice. **1**, **3**, **7** and **0** give the straight views. With the mouse: the right button dragged turns, the middle one moves, the wheel zooms.

Caveats

- Editing a **G** mesh first makes a copy of it as a model; the game's code is not rewritten. The status bar tells the name of the copy.
- Edits that would pass the format limit of 65535 vertices or 65535 triangles are refused and undone.

The preview and cart_meshes

The model you edit is drawn lit and with its textures from the sheet of the cartridge, so you see it as the game will. On the edit page only the edges of the faces that look at the camera are drawn, and only when the mesh has 3000 triangles or fewer.

How the game's meshes arrive

The G meshes come from the kernel function `cart_meshes(path)`. It runs the game's code **apart**, in a Lua state of its own:

- it runs the file's body, then `_init`, `_update` and `_draw` once;
- there is an instruction limit; files, screen and sound do nothing;
- it collects what the code builds with `mesh()`, `mesh_sphere()` and `mesh_cube()`;
- the name of a mesh is the name of the variable that holds it: `M.ship` gives `ship`.

Good to know

- A mesh built only in a later frame, or only after some input, may not be found.
- If the code stops with an error, bm Mesh still shows what was found and tells you the error.
- A G mesh that equals one of the C meshes is not listed twice.

Saving

- **Ctrl+S** saves. **Ctrl+Shift+S** asks for a name (8 letters, saved in `/bme` with the extension `.BME`).
- The menu (Esc) has **Save**, **Save as...**, **Try the game (F5)**, **Import a model...**, **Export to /bmm...** and **Exit bm Mesh**. A mark `*` after the file name at the top means unsaved changes.
- Only the MESH and ANIM sections and the bm Mesh block in the code change. Sprite sheet, map, cover, sound and the rest of the code stay exactly as they were.
- A model with no vertices or no faces is left out of the file: the message says which one.
- A cartridge holds at most 256 models.
- If you opened a game, the first save asks for the editable copy and from then on the file is the `.BME`. To get a game again, build it from the project in the SDK.
- Exit and Open ask you to choose the same entry twice when there are unsaved changes.

Import a .glb

The SD card has two folders for models. `/glb` holds `.glb` files that you copy there; bm Mesh only reads them. `/bmm` holds the converted models (`.bmm`); bm Mesh writes them there. All the names are in lower case.

1. Open a cartridge or a project. Press **i** on the list page, or **Ctrl+I** on any page, or choose **Import a model...** in the menu. On the pad, open the commands with **X** and choose it.
2. The list shows `/bmm` and `/glb`. A name marked `.glb: converted to /bmm` has no `.bmm` yet.
3. Choose a `.bmm` to read it. Choose a `.glb` to convert it: bm Mesh writes `/bmm/NAME.bmm` and opens the model on the edit page.
4. If the name has more than 8 characters, or `/bmm` has the name already, bm Mesh asks for a new name (1 to 8 of a-z, 0-9, `_` and `-`).
5. The status line shows the vertices and the triangles. If the file is not good, it shows the message of the console, for example the format limit of 65535 vertices or triangles.

A `.glb` gives flat colours. The texture of a `.bmm` stays in that file: the faces that used it turn grey, because the project has its own sheet. A `.bmm` with a skeleton keeps it. A `.bmm` with several models gives the first one.

Ctrl+S saves the model in the project, as for any model. **x** on the list page, **Ctrl+E**, or **Export to /bmm...** in the menu writes the mesh to `/bmm/NAME.bmm`; if the file is there, bm Mesh asks before it replaces it.

Tutorials

TRY IT
Do the three tutorials in order with a cartridge of yours that has a model, or with a game that builds its meshes in code.

Move a vertex

1. Start bm Mesh from the **Dev** tab (or **X** on the cover of a game, **Open in bm Mesh**). Choose the cartridge in the file chooser with up and down and Enter.
2. You are on the list page. Choose a mesh with up and down: it turns in the preview.
3. Press **Enter** to edit it. If the mesh is a G mesh, the first change makes a copy as a model.
4. Look at the top line. If it does not say `VERTICES`, press **Tab** until it does.
5. Move the pointer with the arrows to a vertex until it gets a circle around it.
6. Press **Enter**. The top line says `VERTICES: 1 chosen`.
7. Press **g**. The top line shows `MOVE`. Press the **up** arrow once: the vertex moves up by the step.
8. Press **Enter**. The status bar says `move ...: 1 vertices`.
9. Press **Ctrl+S**. You see `saved` with the file name.

If it does not go well: **Esc** while the tool is open puts the vertex back; **Ctrl+Z** after confirming undoes it.

TRY IT
Press **g** again, then **x**, then the right arrow a few times, then **Enter**. The vertex moves only along x.

Fix a face

On the edit page, when the mesh has 3000 triangles or fewer, only the faces that look at the camera have their edges drawn. If a face has no edges from any side, either it is turned the wrong way or it is missing.

The face is turned the wrong way

1. On the edit page, press **Tab** until the top line says `FACES`.
2. Move the pointer on the face and press **Enter**. The face is outlined and the top line says `FACES: 1 chosen`.
3. Press **i**. You see `1 faces turned over`.
4. Press **q** or **e** to turn the view and check it.

The face is missing

1. Press **Tab** until the top line says `VERTICES`.
2. Choose three (or four) vertices with **space**, one after the other, in the order around the face.
3. Press **j**. You see `a new face (1 triangles)`, or 2 for four vertices. The face takes the current colour; **o** and **p** change colours.
4. If it still looks wrong, choose the face and press **i**.

If you see `choose 3 or 4 vertices (in order) for a new face`, the number of chosen vertices is wrong.

Check the model in a game

1. Save with **Ctrl+S**. Remember the name the model has on the list (for a copy of a G mesh, the status bar told you).
2. In the game's code use the name: for a model, `model("name")`.
3. Press **F5**. bm Mesh saves if needed and runs the game.
4. Look at the model in the game. Then come back to the editor: you return to the same page.
5. If something is wrong, edit again and press F5 again.

COPY AND PASTE

```
local m = model("ship")
draw3d(m, 0, 0, 0, 0, 0, 0, 1, 0)
```

To leave the game and come back to bm Mesh, press Start + Select on the pad or Ctrl+Esc on the keyboard; the editor returns to the same page.

Tips and limits

Tips

- Use **1**, **3** and **7** to check a move from the front, the side and the top.
- Use a small step (`,`) to align a vertex, a larger one (`.`) to place a whole part.
- Use **M** to build a symmetrical model: make half, then copy it to the other side.
- Use **K** after joining parts, so the faces around look smooth.
- Save often: the editor remembers up to 100 steps of undo (fewer on a very big model: the steps may take 4 MB together, and at least 2 are kept), and opening or renaming clears them.

Limits

- The format limit of a mesh is 65535 vertices and 65535 triangles. The edit page shows the count of vertices against it.
- At most 256 models in a cartridge.
- A G mesh cannot be renamed or deleted: change the code with bm Code.
- A copy as code does not keep the skeleton.
- The humanoid base (F3 on the list, **New model: humanoid with a skeleton**) has 17 boxes, 136 vertices, 204 triangles and 16 bones. It has no animation: make one in bm Animator.
- The assistant (F6) adds a model with no skeleton.
- **[TO FILL: the size of a model that the Pi Zero W draws smoothly]**

Open points

- **The screen and the controls**: picture to take: shots/editor-mesh-list.png
- **The screen and the controls**: picture to take: shots/editor-mesh-edit.png
- **The screen and the controls**: the pad controls are written from the code; not tried on the PowKiddy RGB30
- **Tips and limits**: the size of a model that the Pi Zero W draws smoothly